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REVIEW ARTICLE

PID Control for Greenhouse Climate Regulation: A Review

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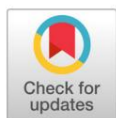
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Abstract

The increasing demand for sustainable agricultural practices has spurred a significant interest in optimizing greenhouse climate regulation to enhance crop yield, quality, and energy efficiency. Proportional-Integral-Derivative (PID) control, a widely used feedback control technique, has emerged as a promising solution for maintaining optimal greenhouse conditions. This paper presents a comprehensive review of the application of PID control in greenhouse climate regulation. A detailed analysis of the PID control algorithm's components, namely Proportional, Integral, and Derivative terms, highlights their respective roles in achieving precise regulation of temperature, humidity, CO₂ levels, and ventilation. Moreover, the review explores the challenges associated with parameter tuning and controller stability in greenhouse settings. The review discusses sensor technologies, and communication protocols, which enable real-time data feedback and enhance the performance of PID controllers. Furthermore, the review presents a comprehensive analysis of recent studies and field applications showcasing the effectiveness of PID control in greenhouse. Comparative studies with other control strategies are also discussed, highlighting the advantages and limitations of PID control in addressing varying climate conditions and crop-specific requirements. Additionally, recent advancements in machine learning-based PID tuning techniques are explored, offering promising alternatives for optimizing greenhouse climate management. The review shows that PID control is a promising approach for greenhouse climate regulation. However, there are still some challenges that need to be addressed, such as the uncertainty of the greenhouse climate model and the non-linearity of the greenhouse system. Future research should focus on addressing these challenges and developing more advanced PID control techniques for greenhouse applications.

Keywords: Smart farm, greenhouse, climate control, PID control, plant growth



DOI: <https://doi.org/10.12972/pastj.20230008>

Received: June 09, 2023

Revised: June 21, 2023

Accepted: June 21, 2023

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introduction

With the global population projected to reach 8.6 billion by 2030 and 9.8 billion by 2050, demand for food and crop production is rapidly increasing (Sagheer et al., 2021). Meeting this demand will require a significant increase in arable land by 42% and water resources by 120% by 2050 (Sagheer et al., 2021; Godfray et al., 2019). Greenhouse farming is emerging as an optimal solution for increasing food and crop production to address water shortages, labor shortages, and other challenges faced by traditional open-field agriculture.

Greenhouse cultivation has gained significant attention in modern agriculture and has received increasing attention in recent years (Choab et al., 2019; Rodríguez et al., 2015). Unlike traditional field production, greenhouse cultivation offers distinct advantages by providing a controlled environment that improves crop yield and quality (Su et al., 2020). The greenhouse structure acts as a protective shield for plants from adverse weather conditions while efficiently utilizing solar energy. The greenhouse environment regulates critical factors such as temperature, humidity, and light intensity to create optimal conditions for plant growth and development. Implementing a greenhouse for cultivation requires the use of various environmental control devices. However, efficient management and monitoring of various environmental parameters is crucial (Postolache et al., 2012). In the past, growers relied on manual observation, requiring significant manpower to maintain optimal growing conditions. However, the advent of smart agriculture has revolutionized greenhouse cultivation (Lytos et al., 2020; López-Cruz et al., 2018).

Designing a greenhouse control system presents several challenges: (1) the complex dynamics of heat and mass exchange within the greenhouse, (2) the significant impact of the internal environment on crop growth, and (3) significant fluctuations in the control process due to unpredictable weather. Therefore, effective greenhouse environmental design focuses on two key aspects: accurately modeling the greenhouse environment and addressing its inherent uncertainties.

Recently, various advanced control techniques and strategies for greenhouse environmental control have been proposed. These include predictive control (Blasco et al., 2007), adaptive control (Su et al., 2017; Hu et al., 2011), nonlinear feedback control (Su et al., 2017; Hu et al., 2011), fuzzy control (Mohamed and Hameed, 2018; Heidari and Khodadadi, 2017), robust control (Bennis et al., 2008; Endo et al., 2022), and optimal control (Wu et al., 2019; Sciarretta et al., 2015). These diverse approaches have contributed to improving the precision of greenhouse environmental control by considering the complex dynamics, uncertainties, and specific requirements of the greenhouse environment. These studies provide important insights and technological advancements for achieving more accurate and efficient temperature control in greenhouse cultivation. While these approaches are useful for practical greenhouse production, many are theoretically complex or difficult to implement in practice. Researchers aim to improve the precision of greenhouse environmental control using the PID algorithm by addressing these limitations through adaptive algorithms, robust control strategies, advanced tuning methods, multivariable control approaches, and energy-efficient optimization techniques. These efforts contribute to the development of more effective control systems that maximize crop productivity while minimizing energy consumption and environmental impact. As a result, most controller designs for greenhouse environmental control still rely on the conventional proportional, integral, and derivative (PID) controller. PID controllers are preferred due to their compact structure, ease of implementation, and superior performance. PID controllers are widely used in various industrial fields such as process control, motor drives, automotive, flight control, and instrumentation, and approximately 95% of control controllers utilize the PID structure (Zheng et al., 2012). In smart farms operated with energy and environmental technologies centered on greenhouses, the element technologies of greenhouse complex environmental control systems are being utilized as greenhouse environmental sensing and control equipment, and the importance of developing remote control technologies utilizing IoT and control theory is increasing (Lim et al., 2023).

The purpose of this study is to conduct a comprehensive review of PID control for greenhouse environmental management. This study aims to examine the effectiveness, limitations, and advancements of PID control in precisely maintaining temperature, humidity, and other environmental parameters within a greenhouse. By synthesizing existing research, this study aims to provide valuable insights into the application of PID control in greenhouse environments, identify the challenges and limitations faced by PID controllers, and explore potential solutions to overcome these limitations. Furthermore, this study aims to highlight the importance of PID control in optimizing crop productivity, energy efficiency, and environmental sustainability in greenhouse cultivation. Overall, this study is expected to serve as a valuable resource for researchers, engineers, and practitioners involved in greenhouse environmental control, helping them develop and implement more efficient and accurate control strategies.

Greenhouse control

Numerous studies are being conducted on environmental control in greenhouse farming. The primary goals of this research are twofold: first, developing accurate models that capture the complex dynamics of greenhouse environments; and second, designing efficient and flexible controllers capable of regulating microclimate variables of interest. This specific problem has garnered significant attention from researchers worldwide, leading to extensive analyses, experiments, and proposals for various environmental control systems. Manipulating key variables such as temperature, relative humidity (RH), CO2 fertilization, and sunlight is crucial for creating optimal conditions for successful protected agriculture. Researchers have strived to develop models that accurately represent the complex relationships and dynamics within greenhouse environments. These models account for the interactions among various factors, such as irradiance, ventilation, crop growth, and external weather conditions.

Researchers are also focusing on the design and implementation of efficient and flexible controllers. These controllers play a crucial role in regulating and maintaining desired microclimate variables within greenhouses. Because control theory typically requires a mathematical representation of the system, various greenhouse models have been developed to facilitate tuning and simulation of control algorithms. These models range from simple models representing only air temperature to more complex models that incorporate crop responses. Existing greenhouse environmental models are typically built on energy and mass balances. These models provide a framework for understanding the dynamics of heat transfer, ventilation, and other critical processes within the greenhouse environment. Researchers can leverage these models to develop and test control algorithms to optimize the regulation of temperature, humidity, and other variables for efficient and precise greenhouse environmental control. Fig. 1 illustrates various greenhouse environmental control algorithms.

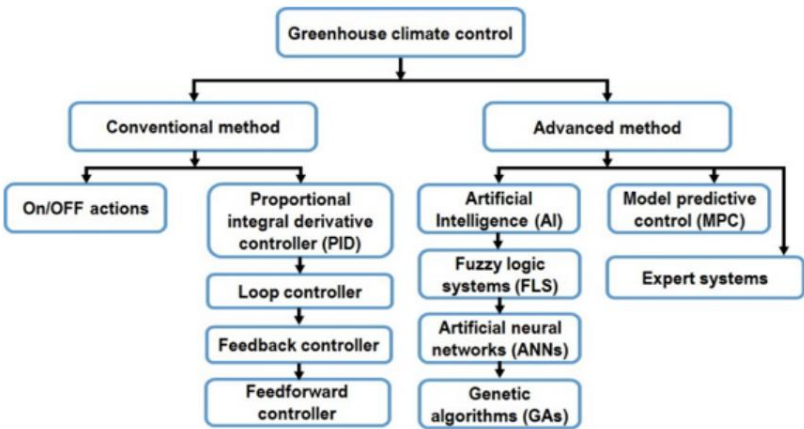


Fig. 1. Classification of greenhouse climate control theories (modified from Duarte-Galvan et al., 2012).

The control methods used in greenhouses are mainly pulse width modulation (PWM) control, PID (Proportional Integral Derivation) control, phase control, and artificial intelligence (Fuzzy) control. Pulse width modulation (PWM) control is a method that controls by adjusting ON/OFF at a high speed, and is used for LED brightness control (dimming) and DC motor speed control. Since the output can obtain a current proportional to the period (duty) value, this method is mainly used in switching power supplies.

Phase control is similar to pulse width modulation (PWM) control, but it is mainly used in incandescent light bulbs, brightness control, vacuum cleaners, and electric drill speed control at home, and it is a method of controlling the speed of the driving motor, but it can only be used when it is AC power and generates noise due to its low frequency. Fuzzy control can control effectively compared to linear controllers due to inherent nonlinearity, and can effectively cope with the control of many nonlinear processes. Unlike PID control, it is a rule-based controller, so it can be changed locally even if some rules are adjusted, but it shows higher performance than existing control methods for changes in parameters (Kabir, 2022).

According to Park and Lee (2022), PID control creates an error using the desired target value and the current system output value and compares it to the target value. It is used as input to the system using a rectifier, integrator, and differentiator. Through this closed loop configuration, only the target value and feedback signal are transmitted. It can be used to reach the desired final output value of the system. PID control can be applied to various systems using some parameters. Because it has the advantage of being capable, it is still used in many fields to this day.

In particular, the control of the facility horticulture greenhouse takes a repetitive action of turning ON/OFF when the sensor value reaches a target value. However, PID control was used as a control method because PID control can prevent this.

PID Control System Overview

Fig. 2 shows a feedback controller using a PID controller, and in the case of temperature control, the manipulation amount given to the control target is a hot wire. In the control target, the control amount fed back to the controller corresponds to the temperature at the time measured by the sensor.

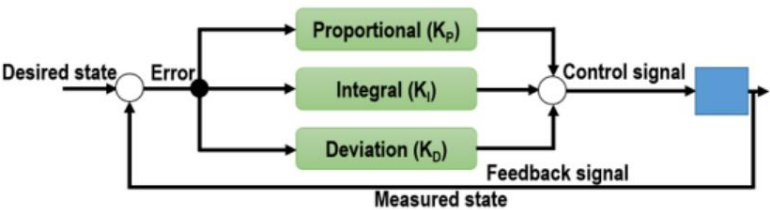


Fig. 2. Block diagram of a PID control (modified from Bansal, 2009).

Lim et al. (2023) stated that PID temperature control is proportional, differential, and integral control, and it is a proportional action that controls responsiveness and a vibration control. It has the advantage of being able to respond with less deviation and higher adaptability by combining differential action and integral action that eliminates deviation in action. He said that he was there.

The PID controller must control the control amount to respond to the target value quickly and stably by combining P-action, I-action, and D-action.

Fig. 3 shows overshoot and hunting that occur during control.

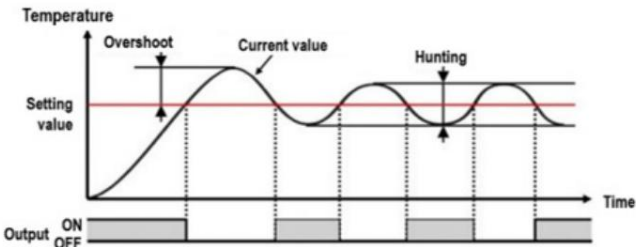


Fig. 3. Over-shoot and hunting of a PID controller (modified from Park and Lee, 2022).

P motion (Proportion)

The operation that is proportional to the deviation, that is, the value obtained by subtracting the control amount from the target value, is called P operation. The operation amount Mp of P operation is,

$$M_p = \frac{e}{P} + M_y$$
 (1)

$$M_p = \frac{100}{P_b} e + M_y$$
 (2)

Here, Kp is the proportional gain, Pb is the proportional band, and represents the intensity of the P action. e is the deviation, and My is the deviation when it is normal. Indicates the amount of manipulation.

As the proportional range narrows, the responsiveness improves, but as shown in Fig. 3, overshoot (a phenomenon in which the control amount exceeds the target value) and hunting (a phenomenon in which the control amount exceeds the target value) occur.

It causes a phenomenon in which the control amount fluctuates, and as it widens, overshoot and hunting decrease, but responsiveness deteriorates. Therefore, the purpose and

An appropriate P_b value must be determined according to the characteristics.

I Action (Integral)

When the control reaches a stable state with only the P operation, the control amount has a normal deviation. This is called an offset, and the deviation

The action that eliminates this offset by integrating over time is called the I action. The manipulation amount M_i of the I action is,

$$M_i = \frac{100}{P_p} \cdot \frac{1}{T_i} \cdot y_{ed} t \quad (3)$$

Here, T_i is the integration time.

If the integration time is long, it takes time to remove the offset caused by the P operation. However, if it is short, the period is long due to the I operation.

It causes tingling.

D action (Differential)

The manipulation amount M_d of the D movement is,

$$M_d = \frac{100}{P_p} T_b \frac{dy}{dt} \quad (4)$$

Here, T_d is the differentiation time. The D operation functions to compensate for the overshoot.

Differences between control operations

Therefore, the input/output relationship of the controller can be expressed as follows.

$$\ddot{y} = K_p \left(e + \frac{1}{T_i} \int e dt + T_b \frac{de}{dt} \right) \quad (5)$$

Where, m = output, e = error, K_p = specific gain, T_i = integration time, T_d = differentiation time.

Although there is a steady-state error in both, as can be seen in Fig. 4, when comparing P control and PD control, the overshoot of PD control is relatively reduced.

You can see the sound. In the case of PID control, the overshoot is similar to that of P control, but the period is longer. As a result, PID control

The point is that the normal state error has disappeared.

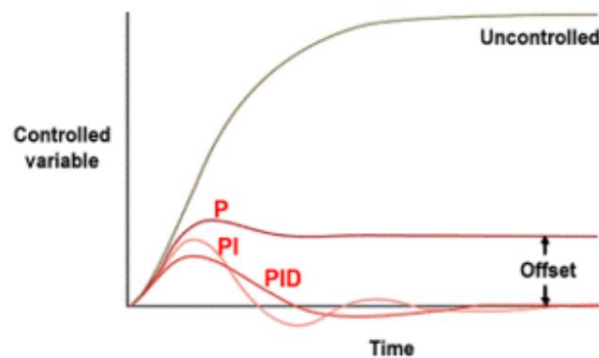


Fig. 4. Comparison between controllers (modified from <https://www.isa.org/>).

Setting the optimal coefficients

There is a problem in PID control that the control performance can vary greatly depending on the parameters. This problem is something that the system operator Finding optimal PID parameters requires the hassle of multiple tests. A method for determining optimal coefficients addresses this issue. Among them, the parameter automatic tuning method developed by Ziegler and Nichols is still widely used and is available in most equipment. The PID automatic tuning method used follows the Ziegler-Nichols Method (Ziegler and Nichols, 1993). This method uses an open loop. The optimal PID control automatic tuning parameters are selected by utilizing the slope and delay time according to the desired step response.

According to the law proposed by Ziegler & Nichole, the proportional gain K_p , the integration time T_i , and the differentiation time T_d are the waste time L of the step response, and the maximum The slope is obtained from the equation below.

$$K_p = \frac{1,2}{RL} T_i = 2 L \quad T_d = 0,5 L \tag{6}$$

In Ziegler & Nichole's step response method, the control target is approximated as [wasted time + 1]. In addition, in the case of digital control, the computational process The time taken for the Rie or sampling cycle is considered as part of the control target waste time.

According to Ziegler & Nichole's law, the final output value varies greatly depending on the values of each parameter variable K_p , K_i , and K_d . Therefore, the parameter size Considering the performance change according to the machine, the parameters of the PID controller must be selected with appropriate values so that the user can use the selected parameters. You can achieve the desired performance (Park and Lee, 2022).

Application of PID control in greenhouses

PID control is widely used in greenhouses to effectively manage and regulate environmental conditions. Proportional, integral, and derivative control Using a proportional component, the PID controller maintains accurate and stable levels of temperature, humidity, and other important parameters. The cow causes the system to respond proportionally to deviations from the set point, and the integral component reduces the steady-state error over time. It helps in optimizing crop growth through PID control, and the differential components predict and respond to sudden changes. , prevent heat stress, minimize energy consumption, and create a favorable environment for crops at each growth stage, ultimately resulting in increased yield. This can lead to improved acidity and increased crop yield. Fig. 5 shows a typical greenhouse environment control scenario.

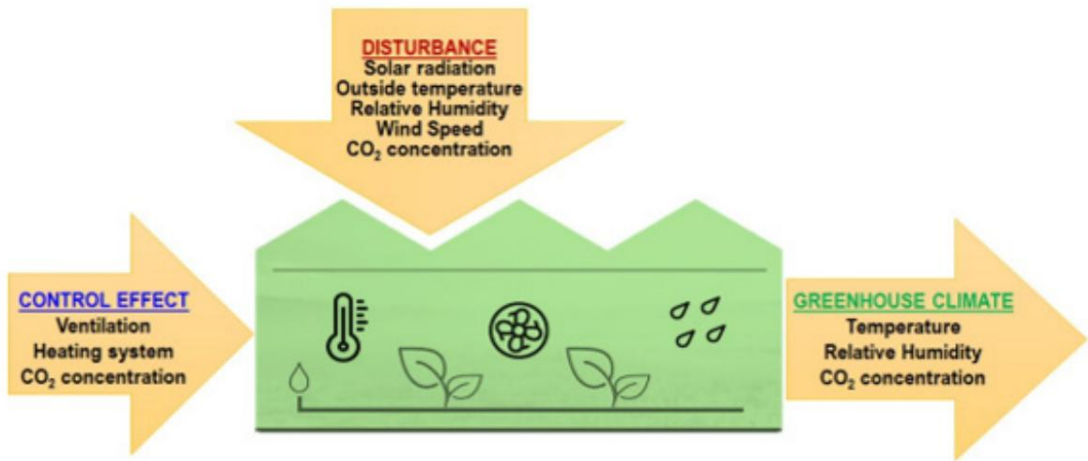


Fig. 5. Greenhouse climate control scenario (modified from El Ghoumari et al., 2005).

Temperature and humidity control in greenhouses

Effective temperature and humidity control in greenhouses is crucial for ensuring optimal crop growth and productivity. To address the challenges of nonlinear and time-varying greenhouse environments, various control approaches have been proposed, including traditional PID controllers using intelligent algorithms, robust controllers based on quantitative feedback theory (QFT), and feedback feedforward systems using fuzzy logic and PID controllers. These innovative control methods aim to improve crop development and production in greenhouses by mitigating rapid environmental changes, stabilizing system response, and enhancing control precision. Wei (2022) developed an intelligent greenhouse temperature control system using a single-chip microcomputer. This system utilizes a PID algorithm to smoothly adjust greenhouse temperature and enable automatic monitoring and control. This innovative solution effectively addresses issues such as low automation, excessive manual labor, and poor control precision in crop cultivation environments. Simulation and field tests demonstrated that the system improved temperature and humidity control accuracy by 10.8%. Heidari and Khodadadi (2017) designed a small greenhouse and implemented a control system using both a fuzzy logic-based PID controller and a self-tuning PID controller. They performed simulations to evaluate the performance of these controllers. This study focused on controlling humidity and temperature within the greenhouse. Comparing the results with a standard PID controller, the proposed architecture demonstrated superior performance in improving greenhouse environmental conditions. This was achieved by reducing rise and stabilization times and minimizing overall overshoot. Wang et al. (2015) proposed an adaptive PID control scheme for greenhouse environmental control using a single neuron approach. They utilized the Hebb learning algorithm to tune the controller parameters. This scheme is easy to implement because it leverages the adaptive, self-organizing, and self-learning capabilities of neurons. The researchers successfully applied this adaptive PID control scheme to greenhouse environmental control problems and demonstrated superior closed-loop performance compared to conventional PID schemes.

Su et al. (2020) introduced a self-tuning parameter-dependent (STPID) control method to improve controller reliability. This method converts three environmental outputs—indoor temperature, humidity, and CO₂ concentration—into four equivalent unit outputs, each controlled by a separate PID controller. To achieve the desired control performance, the Levenberg-Marquart (LM) algorithm is used to adjust the PID parameters. Gurban and Andreescu (2014) developed a solution using a modified Smith predictor to control greenhouse temperature and humidity with decoupling. The system's performance was tested with a 50% variation from nominal conditions to assess robustness. Simulation results compared the control system performance of the temperature and humidity loop using the modified Smith predictor and a PID controller tuned with a genetic algorithm. This study concluded that the modified Smith predictor is a promising and effective solution for greenhouse environmental control systems. Grados et al. (2020) presented a comprehensive analysis of an ideal artificial microclimate system. This system operates independently of external weather conditions, enabling efficient and systematic control. It automates temperature and humidity control using PID-type control, eliminating the need for continuous monitoring. The primary goal is to protect plants and animals from temperature-related threats by creating artificial habitats that maximize their reproduction in diverse environments. This innovative approach promotes biodiversity and conservation efforts by protecting and supporting species that cannot tolerate extreme temperatures. Gao et al. (2018) focused on addressing the challenges posed by nonlinear and time-varying temperature control systems in greenhouses. To overcome this complexity, the researchers proposed integrating conventional PID control with intelligent algorithms. After analyzing the principles and characteristics of PID control, they designed a temperature PID controller for greenhouses using Kalman filtering to improve control accuracy. MATLAB simulations demonstrate the rapid response and efficient performance of the developed temperature controller system. The results significantly improved control efficiency, suggesting a promising solution for greenhouse temperature control. Nunezz and Corzo (2019) studied the damage caused by sudden environmental changes in a naturally ventilated greenhouse. To address this issue, they proposed a robust controller design based on quantitative feedback theory (QFT) with a Smith predictor structure for a dead-time system. This design provided high stability by considering gain margin, phase margin, and disturbance rejection of the system output. They compared the QFT controller with a PID controller based on performance indicators, considering transient response and errors under changing conditions. The results demonstrated that the QFT controller outperformed the PID controller by demonstrating improved dynamic response and control effectiveness. Ahmed et al.

al. (2020) introduced a feedback-feedforward approach for linearization and decoupling of a greenhouse environmental control system. They focused on ventilation, cooling, and humidification operations. Combining controllers including feedback-feedforward, fuzzy logic, and PID controllers, the researchers created an effective and robust control system. Applying this approach, they successfully controlled the temperature and humidity inside the greenhouse, achieving improved environmental control for optimal crop growth.

Effective temperature and humidity control in greenhouses is crucial for optimal crop growth. Stability and precision can be improved through various technological approaches, including PID control using intelligent algorithms, robust QFT-based controllers, and feedback feedforward systems using fuzzy logic and PID controllers. These advancements lead to improved crop yields, reduced manual labor, and increased economic benefits, supporting sustainable and efficient agricultural practices.

Greenhouse lighting control

Light plays a crucial role in the growth of crops in greenhouses. The amount of light a plant receives directly affects its growth rate and lifespan. Light energy is used for photosynthesis, a fundamental metabolic process in plants (Darko et al., 2014). Assessing the impact of light on plant growth requires considering three key aspects: light quality, intensity, and photoperiod (day length). All three light characteristics significantly influence the growth and development of food crops. Therefore, controlling light is an urgent task for optimal food production.

A recent study by Contreras and Begovichan (2022) designed and implemented an automated lighting control system in an indoor greenhouse. This control system utilized a windup-resistant PID controller tuned using a technique called "particle swarm optimization" (PSO). Furthermore, an intuitive user interface was developed, allowing users to monitor illuminance levels and daylight integral (DLI) in real time. The prototype system operated continuously in the greenhouse for 24 hours. The lighting control function was activated at 6:00 AM and deactivated at 8:00 PM daily. This schedule ensured that plants received the light necessary for growth during the specified period.

Mahdavian and Wattanapongsakorn (2013) conducted a study on multi-objective optimization of a PID controller in a greenhouse electric lighting control system. They investigated the electric energy market and optimized a control system for greenhouse lighting. Considering real-time pricing (RTP) in a smart grid, they successfully determined PID controller coefficients to improve the performance of an artificial lighting system in a greenhouse application. Design and simulation results showed significant improvement in the performance of the control system, demonstrating the effectiveness of the approach. Mahdavian and Wattanapongsakorn (2014) focused on optimizing a proportional-integral-derivative (PID) controller in a greenhouse lighting control system. Their approach involved tuning the coefficients of the PID controller. The use of multi-objective optimization methods in various research and engineering scenarios has been a topic of discussion among researchers and engineers. Simulation results confirmed that the output response can be achieved by integrating a larger number of independent objective functions for an improved system. This approach is considered promising for improving the performance of greenhouse lighting control systems. Belkadi et al. (2019) proposed an IoT-based lighting control system for greenhouses. They demonstrated that monitoring and controlling lighting can significantly improve greenhouse efficiency, both in terms of energy consumption and agricultural product quality. Using a PID controller that adjusts lighting based on sensor data, they achieved location-dependent average illuminance levels. As a result, they achieved a significant 12% reduction in energy consumption and operating time per zone, and a 58% reduction compared to conventional control methods.

Implementing lighting control systems in greenhouses using technologies such as IoT-based solutions and PID controllers can yield promising technical results. Lighting monitoring and control have been proven to improve greenhouse efficiency, crop quality, and overall greenhouse performance. These advancements pave the way for sustainable and economically viable greenhouse farming practices.

Controlling air circulation and ventilation in greenhouses

Air circulation and ventilation are crucial elements of greenhouse management, as they play a crucial role in creating and maintaining an environment conducive to crop growth. Appropriate air movement and ventilation control provide optimal distributions of temperature, humidity, and carbon dioxide, which directly impact crop health and productivity. Numerous studies have developed greenhouse ventilation control systems using models based on physical phenomena within the greenhouse (Han et al., 2019; Hong and Lee, 2014). Similarly, some researchers have explored similar approaches that use external factors, such as weather and related environmental data, to guide ventilation adjustments (Francik and Kurpaska, 2020). These methods effectively optimize greenhouse conditions by utilizing control logic based on physical energy conservation or empirical modeling.

Montoya-Ríos et al. (2020) developed a control strategy for greenhouse environmental control, which included a proportional-integral (PI) controller combined with a feedforward compensator to improve performance in the face of external disturbances such as outdoor temperature, irradiance, and wind speed. To address the complexity of greenhouse process dynamics and the mathematically complex physical nonlinear models, the researchers proposed a system identification methodology. This methodology enabled the generation of simpler models, such as high-order polynomials and low-order transfer functions. Using these simplified models, they established an easier procedure for tuning the PI controller parameters and deriving the equations for the feedforward compensator. Hu et al. (2011) investigated the tuning of two PID loops in a MIMO process controller for greenhouse environmental control. They employed an evolutionary algorithm (EA) and several performance measures, including static-dynamic specifications and smooth process control. The results demonstrated that this method was effective in achieving excellent control performance with small overshoot, fast settling time, fast rise time, and reduced steady-state error. The gains obtained in this way were applied to a PID controller to demonstrate the usefulness and practicality of the approach for greenhouse environmental control. Jung et al. (2020) introduced a novel ventilation control logic for a multi-window greenhouse used for strawberry production. They utilized an output feedback neural network (OFNN) prediction model with 15 inputs, demonstrating impressive accuracy with an R2 value of 0.94. Furthermore, the researchers proposed an optimization method based on the OFNN algorithm to effectively manage six window opening operations.

Chouchaine et al. (2011) proposed a control strategy for complex, nonlinear systems using a parallel distributed compensation (PDC) controller. This solution addresses the stability problem of Takagi-Sugeno discrete systems with numerous rules. The PDC controller integrates a classical controller (e.g., PI, PID, or RST) with a pole placement strategy for each rule to ensure stability. A fuzzy controller combines a multi-control approach with classical controller performance to achieve robust nonlinear control suitable for time-varying systems. This method was applied to a small greenhouse to regulate temperature through ventilation rate changes, demonstrating the effectiveness of this approach in controlling nonlinear, complex systems. Susanto et al. (2018) designed a PLC-based control system including a PID controller for a rotating fixture. The stability of the system was analyzed using the Routh-Hurwitz criterion and Nyquist plot. Premkumara and Manikandan (2018) investigated the stability and robustness of a PID controller tuned with ANFIS for a brushless DC motor. They performed a robustness analysis by varying the motor inertia, flux, and resistance from 50% to 200%. They also performed a stability analysis using Nyquist plots.

A PID control system continuously monitors environmental parameters such as temperature, humidity, and CO2 levels and uses a combination of proportional, integral, and derivative actions to control these variables within a desired set point. PID control is a reliable and technically robust approach to controlling air circulation and ventilation in greenhouses. The ability to maintain a consistent and controlled environment ensures optimal conditions for plant growth, ultimately contributing to increased agricultural productivity and sustainable farming practices.

Advantages and Limitations of PID Control in Greenhouse Environmental Control

Greenhouse environmental control plays a pivotal role in optimizing crop growth and productivity by creating an ideal environment that mimics natural conditions. PID control is a widely adopted control strategy for greenhouse environmental regulation. PID control offers various advantages, including robustness, ease of implementation, and real-time adjustments, making it widely used in greenhouse automation. However, like other control methods, PID control has limitations. This article introduces the advantages and limitations of PID control in greenhouse environmental control, as well as technical insights.

Advantages: Accuracy: PID controllers can be used to achieve a high level of accuracy when controlling greenhouse environmental variables. This allows for plant We can guarantee that it can grow under these ideal conditions.

Robustness: PID controllers are relatively robust and can be used to control a wide range of greenhouse environmental variables. They are also resistant to noise and disturbances. It has relatively little influence on people.

Ease of Implementation: PID controllers are relatively easy to implement and can be used with a variety of greenhouse environmental control systems. Flexibility: PID controllers can be customized to meet the specific needs of a greenhouse.

Limitations: Complexity: PID controllers can be complex to tune and may require the expertise of a qualified engineer. Cost: PID controllers can be relatively expensive, especially for large greenhouses. Not suitable for all applications: PID controllers may not be suitable for all greenhouse environmental control applications. In addition to the above, some technical considerations for PID control in greenhouse environmental control include: The time delay between the measurement of the greenhouse environmental variable and the actuator response must be taken into account when tuning the PID controller. The nonlinearity of the greenhouse environmental system must also be considered when tuning the PID controller. Perturbations that can affect the greenhouse environmental system, such as changes in solar radiation or wind speed, must also be considered when tuning the PID controller.

Future Trends and Summary

Recent technological advancements have introduced smart PID controllers and hybrid PID controllers, which have proven to be more effective and cost-effective than conventional PID controllers. Future trends in PID control for greenhouse environmental control are driven by the need for precise and efficient control in agricultural settings. These trends include the integration of advanced control strategies, multivariable control approaches, and the use of data-driven and machine learning technologies. Leveraging these trends can help PID control increase crop productivity, optimize resource utilization, and contribute to sustainable greenhouse operations. Key trends in PID control for greenhouse environmental control include:

Advanced Control Strategies: Future trends in PID control for greenhouse environmental control include the integration of advanced control methods, including machine learning and neural networks. Various control systems and artificial intelligence-based approaches are attracting attention.

Multivariable Control: Multivariable control is gaining popularity in greenhouse environmental control. The focus is shifting from controlling individual variables independently to considering the interdependencies and interactions among various environmental variables. Real-time Monitoring and Sensing: These technologies provide extensive data on environmental variables, crop growth, and system performance. PID control systems can utilize this data to make informed decisions, improve control accuracy, and optimize data utilization.

Integration of IoT and Automation: IoT-enabled devices and sensors collect and transmit real-time data, facilitating remote monitoring and control. Energy Optimization and Sustainability: Energy optimization and sustainability are key trends in greenhouse environmental control. PID control algorithms can be further optimized to minimize energy consumption while maintaining desired environmental conditions. Data-driven and Machine Learning Approaches: The increasing utilization of big data and advancements in machine learning technology are bringing new applications of PID control to greenhouse environmental control. It provides new opportunities.

Cyber-Physical System: Controlling the greenhouse environment through seamless integration of sensors, actuators, controllers, and data analysis. can be optimized. A PID control system integrated with a virtual physical system can improve control accuracy, efficiency, and adaptability.

In summary, future trends in PID control for greenhouse environmental control will revolve around advanced control strategies, multivariable control, real-time monitoring and sensing, IoT and automation, energy optimization and sustainability, data-driven approaches, and the integration of cyber-physical systems. These trends aim to improve control accuracy, adaptability, energy efficiency, and sustainability in greenhouse operations.

conclusion

PID control has been extensively studied and utilized for greenhouse environmental control due to its simple structure, ease of use, and excellent performance. This review highlights the importance of PID control in precisely maintaining temperature, humidity, and other environmental parameters in greenhouses. Research in this area has focused on addressing the complex dynamics of greenhouse environments, the impact of environmental changes on crop growth, and the uncertainties associated with weather fluctuations. While PID control has proven effective, limitations have been identified, such as difficulties in handling nonlinear dynamics, uncertainty, and tuning issues. To overcome these limitations, researchers have proposed various methods, including adaptive algorithms, fuzzy-based PID control, robust control, and advanced tuning techniques. These advancements aim to improve the precision and robustness of PID control in greenhouse environmental control.

This review also highlighted the importance of accurately modeling greenhouse dynamics and integrating PID control with other advanced control strategies, such as predictive control and AI-based approaches. Such integrated approaches have shown promise in improving control performance, energy efficiency, and crop productivity. PID control remains a popular and effective choice for greenhouse environmental control. Ongoing research efforts are focused on addressing the limitations and challenges associated with PID control and exploring innovative approaches to improve control precision and efficiency. With continued advancements in control technologies and modeling methodologies, the future of greenhouse environmental control holds great potential to optimize crop growth, resource utilization, and environmental sustainability.

Acknowledgements

This study was supported by the Ministry of Agriculture, Food and Rural Affairs (hereinafter referred to as the Ministry of Agriculture, Food and Rural Affairs).¹ Through the 'Agricultural and Food Technology Convergence Creative Talent Training Project' (Project No. 320001-4) This work was carried out with the support of the National Institute of Food Technology Planning and Evaluation (IPET).

References

- Ahmed M, Abdelhafidh M, Nouredine OC. 2014. Climate Control of Greenhouse Using Fuzzy Logic and PID Controllers. *International Journal of Electrical Energy*, 2(3): 232-237.
- Bansal HO. 2009. Tuning of PID Controllers Using Simulink. *International journal of Mathematical modeling, Simulation and Applications*, 2(3): 337-344.
- Belkadi A, Mezghani D, Mami A. 2019. IoT and lighting control for smart greenhouse. Zenodo. DOI: <http://doi.org/10.5281/zenodo.3813434>.
- Bennis N, Duplaix J, Enéa G, Haloua M, Youlal H. (2008). Greenhouse climate modelling and robust control. *Computers and electronics in agriculture*, 61(2): 96-107.
- Contreras C, Begovich O. 2022. Automatic Lighting Control and IoT Monitoring on an Indoor-Greenhouse. In *2022 IEEE 18th International Conference on Automation Science and Engineering (CASE)* (pp. 1232-1237). IEEE.

- Choab N, Allouhi A, El Maakoul A, Kousksou T, Saadeddine S, Jamil A. 2019. Review on greenhouse microclimate and application: Design parameters, thermal modeling and simulation, climate controlling technologies. *Solar Energy*, 191: 109-137.
- Chouchaine A, Feki E, Mami A. 2011. Stabilization using a discrete fuzzy PDC control with PID controllers and pole placement: application to an experimental greenhouse. *Journal of Control Science and Engineering*, 2011: 1-9.
- Darko E, Heydarizadeh P, Schoefs B, Sabzalian MR. 2014. Photosynthesis under artificial light: the shift in primary and secondary metabolism. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 369(1640): 20130243.
- Duarte-Galvan C, Torres-Pacheco I, Guevara-Gonzalez RG, Romero-Troncoso RJ, Contreras-Medina LM, Rios-Alcaraz MA, Millan-Almaraz JR. 2012. Advantages and disadvantages of control theories applied in greenhouse climate control systems. *Spanish Journal of Agricultural Research*, 10(4): 926-938.
- El Ghomari MY, Tantau HJ, Serrano J. 2005. Non-linear constrained MPC: Real-time implementation of greenhouse air temperature control. *Computers and electronics in agriculture*, 49(3): 345-356.
- Endo A, Cartagena O, Ocaranza J, Sáez D, Muñoz C. 2022. Fuzzy and neural prediction intervals for robust control of a greenhouse. In 2022 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE) (pp. 1-8). IEEE.
- Francik S, Kurpaska S. 2020. The use of artificial neural networks for forecasting of air temperature inside a heated foil tunnel. *Sensors*, 20(3): 652.
- Gao Z, He L, Yue X. 2018. Design of PID controller for greenhouse temperature based on Kalman. In *Proceedings of the 3rd International Conference on Intelligent Information Processing* (pp. 1-4).
- Godfray HCJ, Beddington JR, Crute IR, Haddad L, Lawrence D, Muir JF, Pretty J, Robinson S, Thomas SM, Toulmin C. 2010. Food security: the challenge of feeding 9 billion people. *science*, 327(5967): 812-818.
- Grados J, Auccahuasi W, Rubiños S, Teran C, Rafael G, Enoky W. 2020. Prototype Of A PID Type Temperature And Humidity Control System Of An Artificial Habitat For Animals And Plants Of Different Climate. *International Journal of Scientific & Technology Research*, 9(4): VOLUME 9, ISSUE 04, 392-394.
- Gurban EH, Andreescu GD. 2014. Comparison of modified Smith predictor and PID controller tuned by genetic algorithms for greenhouse climate control. In 2014 IEEE 9th IEEE International Symposium on Applied Computational Intelligence and Informatics (SACI) (pp. 79-83). IEEE.
- Han S, Li Y, Zhou B, Liu Z, Feng J, Xiao Y. 2019. An in-situ accelerated experimental testing method for drip irrigation emitter clogging with inferior water. *Agricultural Water Management*, 212: 136-154.
- Hong SW, Lee IB. 2014. Predictive model of micro-environment in a naturally ventilated greenhouse for a model-based control approach. *Journal of Bio-Environment Control*, 23(3): 181-191.
- Heidari M, Khodadadi H. 2017. Climate control of an agricultural greenhouse by using fuzzy logic self-tuning PID approach. In 2017 23rd International Conference on Automation and Computing (ICAC) (pp. 1-6). IEEE.
- Hu H, Xu L, Wei R, Zhu B. 2011. Multi-objective control optimization for greenhouse environment using evolutionary algorithms. *Sensors*, 11(6): 5792-5807.

- Jung DH, Kim HJ, Kim JY, Lee TS, Park SH. 2020. Model predictive control via output feedback neural network for improved multi-window greenhouse ventilation control. *Sensors*, 20(6): 1756.
- Kabir MS, Islam S, Ali M, Chowdhury M, Chung SO, Noh DH. 2022. Environmental sensing and remote communication for smart farming: a review. *Precision Agriculture Science and Technology*, 4(2):81-90.
- Lim JW, Reza N, Chung SO, Lee KY, Lee SY, Lee KN, Lee B. 2023. Application of artificial neural network in smart protected horticulture: A review. *Precision Agriculture Science and Technology*, 5(1): 29:41.
- Lopez-Cruz IL, Fitz-Rodriguez E, Salazar-Moreno R, Rojano-Aguilar A, Kacira M. 2018. Development and analysis of dynamical mathematical models of greenhouse climate: A review. *Eur. J. Hortic. Sci*, 83(5):269-280.
- Lytos A, Lagkas T, Sarigiannidis P, Zervakis M, Livanos G. 2020. Towards smart farming: Systems, frameworks and exploitation of multiple sources. *Computer Networks*, 172: 107147.
- Mahdavian M, Wattanapongsakorn N. 2013. PID controller tuning and optimizing for greenhouse lighting application considering real-time pricing in the smart grid. In *International Computer Science and Engineering Conference (ICSEC)* (pp. 85-90). IEEE.
- Mahdavian M, Wattanapongsakorn N. 2014. Optimizing PID controller tuning for greenhouse lighting control system by varying number of objectives. In *11th International Conference on Electrical Engineering/ Electronics, Computer, Telecommunications and Information Technology (ECTI-CON)* (pp. 1-6). IEEE.
- Mohamed S, Hameed IA. 2018. A GA-based adaptive neuro-fuzzy controller for greenhouse climate control system. *Alexandria Engineering Journal*, 57(2): 773-779.
- Montoya-Rios AP, Garcia-Mañas F, Guzman JL, Rodriguez F. 2020. Simple tuning rules for feedforward compensators applied to greenhouse daytime temperature control using natural ventilation. *Agronomy*, 10(9): 1327.
- Nuñez RRA, Corzo RCL. 2019. Real-time QFT Control for Temperature in Greenhouses. *Maskay*, 9(2): 58-62.
- Park YS, Lee DJ. 2022. Optimal PID Control for Temperature Control of Chiller Equipment. *Journal of the Society of Korea Industrial and Systems Engineering*. 45.3 (2022): 131-138. [in Korean]
- Postolache O, Pereira JM, Girão PS, Monteiro AA. 2012. Greenhouse environment: Air and water monitoring. *Smart Sensing Technology for Agriculture and Environmental Monitoring*, 81-102.
- Premkumar K, Manikandan BV. 2018. Stability and performance analysis of ANFIS tuned PID based speed controller for brushless DC motor. *Current Signal Transduction Therapy*, 13(1): 19-30.
- Sagheer A, Mohammed M, Riad K, Alhajhoj M. 2020. A cloud-based IoT platform for precision control of soilless greenhouse cultivation. *Sensors*, 21(1): 223.
- Sciarretta A, De Nunzio G, Ojeda LL. 2015. Optimal ecodriving control: Energy-efficient driving of road vehicles as an optimal control problem. *IEEE control systems magazine*, 35(5): 71-90.
- Shi P, Luan X, Liu F, Karimi HR. 2012. Kalman filtering on greenhouse climate control. In *Proceedings of the 31st Chinese Control Conference* (pp. 779-784). IEEE.

- Su Y, Yu Q, Zeng L. 2020. Parameter self-tuning PID control for greenhouse climate control problem. *IEEE Access*, 8: 186157-186171.
- Su Y, Xu L, Goodman ED. 2017. Greenhouse climate fuzzy adaptive control considering energy saving. *International Journal of Control, Automation and Systems*, 15: 1936-1948.
- Susanto A, Sampurno S, Suhardjono S. 2019. Design of PLC Based Control System for Rotary Flexible Fixture with PID Compensator. *IPTEK The Journal of Engineering*, 4(3): 26-32.
- Wang W, Xu L, & Hu, H. (2015). Neuron adaptive PID control for greenhouse environment. *Journal of Industrial and Production Engineering*, 32(5), 291-297.
- Wei X. 2022. Intelligent Temperature Control System of Greenhouse Based on STM32 Single Chip Microcomputer. In *Journal of Physics: Conference Series* (Vol. 2254 (1), p. 012046). IOP Publishing.
- Wu Y, Li L, Li S, Wang H, Zhang M, Sun H, Sygrimis N, Li M. 2019. Optimal control algorithm of fertigation system in greenhouse based on EC model. *International Journal of Agricultural and Biological Engineering*, 12(3): 118-125.
- Zeng S, Hu H, Xu L, Li, G. 2012. Nonlinear adaptive PID control for greenhouse environment based on RBF network. *Sensors*, 12(5): 5328-5348.
- Ziegler JG, Nichols NB. 1993. Optimum Settings for Automatic Controllers. *Journal of dynamic systems, measurement, and control*. 115(2B): 220-222.